Automounting NFS Share using autofs

Objective
To automatically mount an NFS shared directory from the server whenever the client accesses it using autofs.
Before configuring autofs, ensure:

- NFS server configuration is completed
- NFS client packages are installed
- Exported directories are accessible from the client

Files used in automount configuration:
1. /etc/auto.master
2. /etc/auto.misc

Steps to configure automount:

STEP 1: -

Search for autofs package –
#yum search autofs

```
[root@localhost ~]# yum search autofs
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscripti
on-manager to register.

Last metadata expiration check: 0:14:04 ago on Tue 12 May 2026 11:40:41 PM IST.
======================== Name Exactly Matched: autofs ========================
autofs.x86_64 : A tool for automatically mounting and unmounting filesystems
======================= Name & Summary Matched: autofs =======================
libsss_autofs.x86_64 : A library to allow communication between Autofs and SSSD
```

STEP 2: -

Install autofs-
#yum install autofs

```
[root@localhost ~]# yum install autofs.x86_64
```

autofs is used to automatically mount remote directories only when they are accessed by the user.

*In this configuration, the client system uses the default automount directory:

/misc

The file:

/etc/auto.master already contains the default entry:

```
#
# Sample auto.master file
# This is a 'master' automounter map and it has the following format:
# mount-point [map-type[,format]:]map [options]
# For details of the format look at auto.master(5).
#
/misc   /etc/auto.misc
```

/misc /etc/auto.misc

**The /misc directory is managed dynamically by autofs and does not need to be manually created by the user. This entry tells autofs to monitor the /misc directory and use the configuration present in: /etc/auto.misc

Step 3: -

The following entry is added in vi /etc/auto.misc

```
#
# This is an automounter map and it has the following format
# key [ -mount-options-separated-by-comma ] location
# Details may be found in the autofs(5) manpage

cd              -fstype=iso9660,ro,nosuid,nodev :/dev/cdrom

# the following entries are samples to pique your imagination
#linux          -ro,soft                ftp.example.org:/pub/linux
#boot           -fstype=ext2            :/dev/hda1
#floppy         -fstype=auto            :/dev/fd0
#floppy         -fstype=ext2            :/dev/fd0
#e2floppy       -fstype=ext2            :/dev/fd0
#jaz            -fstype=ext2            :/dev/sdc1
#removable      -fstype=ext2            :/dev/hdd
localdirnfs   -rw   10.0.0.50:/nfssharedir
```

localdirnfs -rw 10.0.0.50:/nfssharedir

Here:

- localdirnfs represents the client-side automount directory
- 10.0.0.50:/nfssharedir represents the NFS shared directory from the server

Step 4: -

After configuration restart the service -

#systemctl restart autofs

This command reloads the automount configuration and applies the changes made in configuration files.

Step 5: -

Verification-

```
[root@localhost ~]# cd /misc/localdirnfs/
[root@localhost localdirnfs]# df -h
Filesystem             Size  Used Avail Use% Mounted on
devtmpfs               4.0M     0  4.0M   0% /dev
tmpfs                  1.8G     0  1.8G   0% /dev/shm
tmpfs                  727M  9.8M  718M   2% /run
/dev/sdb1               10G  4.7G  5.4G  47% /
tmpfs                  364M  100K  364M   1% /run/user/0
/dev/sr0               9.0G  9.0G     0 100% /mnt
10.0.0.50:/nfssharedir  10G  5.4G  4.7G  54% /misc/localdirnfs
```

When the user accesses:

/misc/localdirnfs

autofs automatically:

1. Reads the mapping from /etc/auto.misc
2. Contacts the NFS server
3. Mounts the remote shared directory dynamically
4. Provides access to the user

Conclusion:

autofs provides dynamic mounting of NFS shared directories. The remote filesystem is mounted only when accessed by the user and unmounted automatically after inactivity, improving system efficiency and reducing unnecessary permanent mounts.